



Plant Management System

Group C

Semi-Automatic Information and Knowledge Systems

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Dataset - Trefles Species Dump



Raw Facts

- 417.000+ species
- 54 features

Why Trefle fits

- Covers taxonomy, morphology, ecology, and cultivation in a single file, giving us 35+ classes and 32+ properties naturally
- Multi-value fields (bloom months, edible parts) map directly to OWL object properties

Datatype Properties	Object Properties
hasScientificName	belongsToFamily
hasMaximumHeightCm	bloomsInMonth
hasPhMinimum	isNativeTo
isEdible	hasGrowthHabit
hasPhMinimum	hasGrowthRate
hasPlantingDaysToHarvest	hasEdiblePart

Few Competency Question Examples

Our Plant Management System should be able to help both customers and the owner of the shop.



Customer perspective:

- I have a low-light apartment and a curious cat, what plants can I buy?
- Which plants tolerate high soil salinity and require low light (≤ 3)?
- What is an easy-to-care-for climbing plant that costs under 30 Euros?
- Which plants have both edible leaves and edible roots?

Shop Owner perspective:

- Which products in our inventory require 'High Humidity' and 'Warm Temperatures', and have been sitting on the shelf for over three months?
- What is our total stock quantity of all plants belonging to the Araceae family/genus?
- Do we have enough (>5) Orchids on stock?

Current Status and Next Steps



Current Status

Topic: Plant Management System

Source: Trefle Dataset

Completed: Initial Ontology Draft

Sample Classes:

Plant, Species, Flower

Example Properties:

belongsToFamily, isEdible

Next Steps

- Build Ontology in Protege
- Define Database Scheme
- Create Mappings & SHACL Validation
- Build a simple Application